

Experiment - 01

- * Aim: Design and implement parallel Breadth first search and depth first search based on existing algorithms using OPENMP. Use a tree or an undirected graph for BFS and DFS.

- * THEORY:

- (i). What is BFS ?

- ° BFS stands for breadth first search. It is a graph traversal algorithm used to explore all the nodes of a graph or a tree systematically, starting from root node or a specified starting point and visiting all the neighbouring nodes at the current depth level before moving on to the next depth level.
- ° The algorithm uses a queue data structure to keep track of the nodes that need to be visited and marks each visited node to avoid processing it again.
- ° The basic idea of BFS algorithm is to visit all the nodes at a given level which ensures that all nodes are visited in breadth first order.

- o BFS is commonly used in many applications, such as finding the shortest path between two nodes, solving puzzles and searching through a tree or graph.

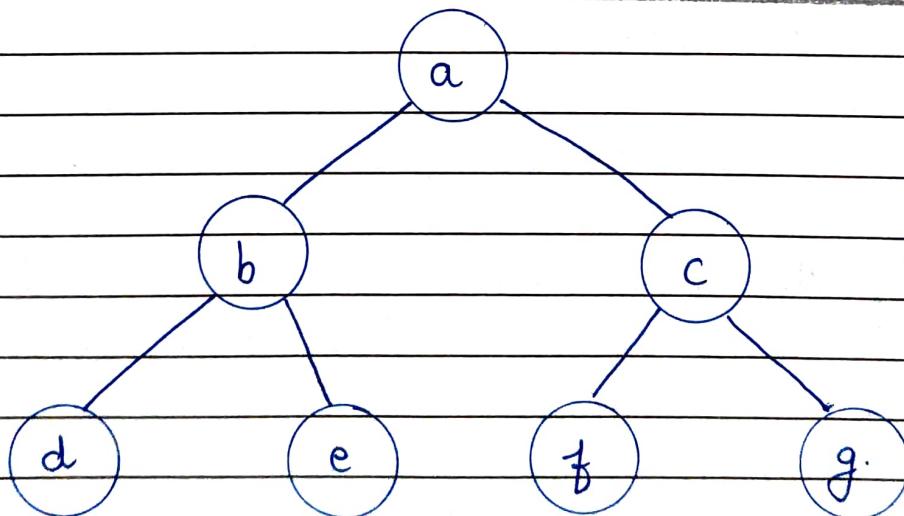
* Example of BFS -

Step 1: Take an empty queue.

Step 2: Select a starting node (visiting a node) and insert it into the queue.

Step 3: Provided that the queue is not empty, extract the node from the queue and insert its child nodes (exploring a node) into the queue.

Step 4: Print the extracted node.



Queue:

						a
--	--	--	--	--	--	---

Print a:

					c	b
--	--	--	--	--	---	---

Print a - insert child nodes.

Print b:

				e	d	c
--	--	--	--	---	---	---

Print c:

			g	f	e	d
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Print c and insert its child nodes.

Print d:

				g	f	e
--	--	--	--	---	---	---

Print e:

					g	f
--	--	--	--	--	---	---

Print f:

						g
--	--	--	--	--	--	---

Print g:

--	--	--	--	--	--	--

Similarly for all consecutive nodes.

(2) CONCEPT OF OPEN MP -

- Open MP (Open multiprocessing) is an application programming interface (API) that supports shared memory parallel programming in C, C++ and Fortran.

It is used to write parallel programs that can run on multicore processors, multiprocessor systems and parallel computing clusters.

- Open MP provides a set of directives and cluster functions that can be inserted into the source code of the program to parallelize its execution.

- These directives are simple and easy to use and they can be applied to loops, sections, functions and other programs constructs. The compiler then generates parallel code that can run on multiple processors concurrently.

- Open MP programs are designed to take advantage of shared memory architecture of modern processors.

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where multiple processor cores can access the same memory. OpenMP uses a Fork-join model of parallel execution, where a master thread forks multiple worker threads to execute a parallel region of code and then waits for all threads to complete before continuing with the sequential part of the code.

- OpenMP is widely used in scientific computing, engineering and other fields that require high performance computing.
- It is supported on most modern compilers and is available on a wide range of platforms, including desktops, servers and supercomputers.

(4) How parallel BFS works -

- Parallel BFS is an algorithm used to explore all nodes of a graph or tree symmetrically in parallel. It is a popular parallel algorithm used for graph traversal in distributed computing, shared memory systems and parallel clusters.
- The parallel BFS algorithm starts by selecting a root node or a specified starting point and then assigning it to a thread or processor in the system. Each thread maintains a local queue of nodes to be visited and marks each visited node to avoid processing it again.

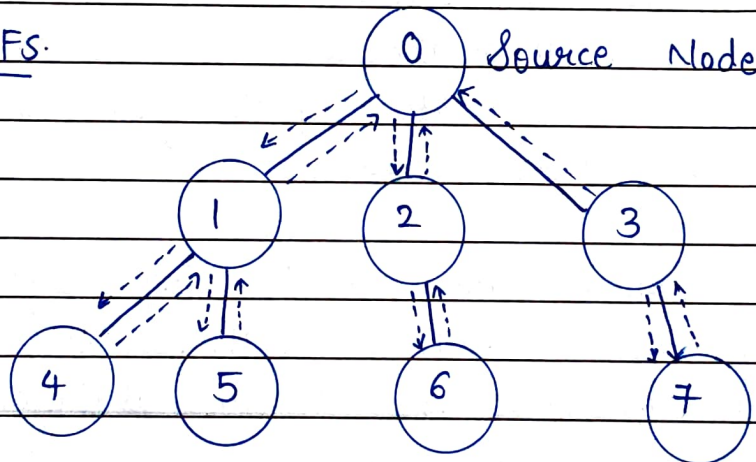
- The parallel BFS algorithm uses two phases: the computation phase and the communication phase. In the computation phase each thread processes the nodes in its local queue, while in the communication phase the threads exchange the nodes that are adjacent to the current level with the other threads or processors.
- The parallel BFS algorithm terminates when all nodes have been visited or when a specified node has been found. The result of the algorithm is the set of visited nodes or the shortest path from the root node to the target node.
- Parallel BFS can be implemented using different parallel programming models such as OPENMP, MPI, cuda, and others. The performance of the algorithm depends on the number of threads or processors used, the size of the graph, and the communication overhead between the threads or processors.

(5). Parallel Depth First Search (DFS) -

- DFS is another graph traversal algorithm that explores all the vertices of a graph or tree by recursively exploring as far as possible along each branch before backtracking

- To implement a parallel version of DFS using OPENMP, we can use a stack data structure that will hold the vertices to be processed
- Each thread will pick a vertex from the stack, process it and add its unvisited neighbours to the stack. This process will continue until the stack is empty.

DFS:



Output - 0, 1, 4, 5, 2, 6, 3, 7

* Conclusion -

Thus we have successfully implemented parallel BFS and DFS using openMP to enhance computational capacity and results of the algorithms.